Full Methods and Results for Experiments 1 to 5 of "How Conflicts of Interest Affect Trust in Science"

Experiment 1

Participants and design

Experiment 1, which asked people about their trust in a study designed to test the efficacy of a new drug, is based on a 2 (*conflict of interest*: conflict vs. no conflict) × 2 (*study outcome*: positive vs. negative) between-subjects design. We aimed to recruit 800 participants based on a Bayes factor design analysis (BFDA, Schönbrodt & Wagenmakers, 2018). In this BFDA, we assumed that a conflict of interest decreases trust by $d = 0.30$ if the results are positive and increases it by $d = 0.30$ if the results are negative. Using these parameters, we ran 25,000 simulations and applied the planned tests on each instance. For the test of the resulting interaction effect (Hypothesis 1), the simulation showed that the likelihood of detecting the effect was close to 95% with a false negative rate of less than 0.3% and the remaining 5% of cases being inconclusive. For the simple effects of conflict of interest (Hypotheses 2 and 3), the sample size ensures a true positive rate close to 80% with a false negative rate of 1.5% and the remaining cases with inconclusive results.

Our sample consisted of 803 participants recruited via Prolific who received £1 as compensation. Participation was conditional on participants being UK residents aged 18 and over. We had to exclude 22 participants who did not provide consent to participate and another 115 who failed an attention check. Originally, we preregistered to replace participants who failed the attention check. However, since the data was already conclusive with the remaining 666 participants and unlikely to change substantially when increasing the sample size to 800 based on an interim design analysis (Stefan et al., 2024), we chose stop data collection.

Procedure

We presented participants the title page of a fictional journal article. This page contained the journal name and issue, study title, and authors names along with an abstract (labelled "summary") and a transparency statement. The fictional study tested a new drug for a fictional disease, in which this new drug was compared to an existing state-of-the art drug in a randomised control trial.

We manipulated conflict of interest via the transparency statement. In the conflict condition, the authors reported that they received funding for their study from the company that had developed the new drug. In in the no conflict condition, the authors stated that they had no conflicts of interest. Study outcome was manipulated via the summary on the title page. Depending on the condition, the new drug or method either significantly outperformed or underperformed the current standard drug. The new drug's performance was manipulated via the proportion of patients free of symptoms after six months (significantly more vs. significantly fewer patients than the current standard drug) while stating that the frequency and severity of side effects did not differ between the two drugs. The study summary ended with a recommendation by the authors that the new drug or method should be adopted (or should not be adopted) as the new standard.

Prior to reading the title page, participants were informed that the journal, study, authors, and disease were fictional, but that we created them to resemble actual scientific journal articles. After reading the title page, we asked them to indicate their agreement with four statements, namely that study results, the authors, the journal that published the study, and the company that developed the new drug (or wastewater treatment method in Experiment 2) can be trusted. For each of the four trust ratings, participants used a 7-point scale (1 – completely disagree to 7 – completely agree). Thus, higher values indicate greater trust. In addition, we asked participants to indicate how likely they thought it would be that an independent team of researchers would be able to replicate the findings of the presented study.

To this end, they entered a percentage value were 0% indicated that participants were certain that the independent researchers would end up with results that supported a different conclusion and 100% indicating certainty that the replication would end up with results supporting the original conclusion. Finally, we included an attention check asking participants to state whether the authors had reported a conflict of interest or not. Participants who failed this attention check were excluded from data analysis.

Results

To test our hypotheses, we employed Bayesian hypothesis testing, using the Bayes factor (BF) as our inference criterion. We treated a Bayes factor (BF) larger than 3 as evidence for an effect, a BF < 1/3 as evidence for its absence, and BFs in between as inconclusive. We preregistered the following analysis plan. First, we would test for an interaction of conflicts of interest and study results on the three focal dependent variables (trust in the study, trust in the authors, and estimated replicability) using Bayesian ANOVAs as implemented in the BayesFactor package for R. We used the default priors for the effect size and tested for the interaction effect by comparing the full model containing both main effects and the interaction to a model containing only the two main effects. In case this analysis would yield evidence for an interaction, we would proceed to test Hypotheses 2 and 3 via simple effects of conflicts of interest using one-tailed Bayesian *t*-tests with a scaling factor of $r = 0.50$ for the Cauchy prior. Finally, in case these *t*-tests yielded evidence against our hypothesis, we would rerun them as two-tailed tests to differentiate between two possible outcomes that a one-tailed *t*-test cannot discern, namely a null effect and an effect in the opposite direction.

Supporting Hypothesis 1, we found clear evidence for an interaction effect on trust in the study results (BF > 100), trust in the authors (BF > 100), and estimated replicability (BF = 17.18). However, inspection of the simple effects showed that the interaction pattern differed partly from the one we predicted. In line with Hypothesis 2, when the results were positive, conflicts of interest led to a substantial decrease in trust on all three focal measures (all BFs >

100). However, when the results were negative, participants did not report greater trust in the study ($BF = 0.10$) nor in the authors ($BF = 0.18$), and there was also evidence against greater estimated replicability ($BF = 0.20$). Rerunning the three tests as two-tailed *t*-tests showed that conflicts of interest had, in fact, no effect on trust ratings and estimated replicability when the results were negative (all BFs < 0.25), leading us to reject Hypothesis 3.

Our exploratory analysis of the two remaining variables, trust in the journal that published the study and trust in the company that had developed the drug showed essentially the same pattern, that is, there was an interaction of the two independent variables (both BFs > 100) due to a conflict of interest decreasing trust with positive results(both BFs > 100) and no change in confidence with negative results (both one-tailed BFs < 0.20, both two-tailed BFs < 0.18).

Finally, for the sake of completeness, we analysed the simple effects of study outcomes. For the condition with conflicts of interest, we used one-tailed t-tests since we expected trust to be greater when the results were negative. This was confirmed for all four trust measures and estimated replicability (all BFs > 100). Expecting that study results would not have a substantial impact in the absence of a conflict of interest, we ran a set of two-tailed *t*-tests for the remaining simple effect of study results. For trust in the study, the authors, and the journal, these tests indicated that study results had no impact on trust (all BFs < 0.18), but for trust in the company ($BF = 0.34$) and estimated replicability ($BF = 0.37$) the result was inconclusive.

The results of Experiment 1 provided partial support for our predictions: Although we found the expected interaction between conflict of interest and study outcome, and the expected loss in trust when conflicts of interest coincided with positive study results, there was conclusive evidence that scientists who publish negative findings do not benefit from a trust boost. In Experiment 2 we tested the possibility that negative associations with

pharmaceutical companies as greedy and immoral masked the expected trust boost for researchers who publish results that run counter to their funder's interests.

Experiment 2

Participants and design

Experiment 2, which asked participants about their trust in a study designed to test the efficacy of a novel method for treating wastewater, was based on the same two-factorial design as Experiment 1. Participants were 798 UK residents aged 18 years or older, recruited via Prolific, who received £0.50 as compensation.[1] Participants in Experiment 1 were prohibited from taking part in Experiment 2. Using resampling of data from Experiment 1 with replacement, we determined a minimum sample size of 400 participants via a BFDA. This sample size guaranteed a chance of over 90% to detect the interaction effect observed in Experiment 1 and a chance exceeding 99% to detect the detrimental effect of conflicts of interest when study results are positive. In the condition with negative study results, the BFDA revealed close to 90% chance to find evidence against an effect of conflicts of interest should this null effect replicate in a different scenario.

After collecting data from the initial sample of 400, some of the focal tests were inconclusive. Thus, we decided – in line with the preregistration – to gather additional data. We stopped at roughly 800 participants when most of the individual hypothesis tests were conclusive, and the overall pattern was unlikely to change further. Similar to Experiment 1, we had to exclude 7 participants who did not provide consent to participate and 102 participants who failed the attention check. This left us with a final sample of 689 participants.

[1] In Experiment 1, we substantially overestimated how long it would take participants to complete the study. We adjusted the compensation in the following experiments, accordingly.

Procedure

The procedure of Experiment 2 is identical to that of Experiment 1 with one exception. We changed the content of the study title and summary, which now described a test of the effectiveness of a new sewage treatment method. This new method was compared to an existing standard method in a randomised control trial. Depending on the study results condition, the new method either led to fewer residual pollutants at lower cost or more residual pollutants while incurring greater costs than the current standard method.

Results

We first tested for the interaction of conflict of interest and study results. In line with Hypothesis 1, we found evidence for an interaction effect on trust in the study, trust in the authors, and estimated replicability (all *BF*s > 100). To disentangle this interaction, we next tested the simple effect of conflicts of interest when the study result was positive. Similar to Experiment 1 and in line with Hypothesis 2, trust decreased in those cases when the authors reported a conflict of interest, and participants estimated a lower likelihood of the study's replicability (all *BF*s > 100). When the study result was negative, there was evidence against the trust boost we predicted in Hypothesis 3 for trust in the study (one-tailed $BF = 0.12$; two tailed $BF = 0.19$) and trust in the authors (one-tailed $BF = 0.09$, two tailed $BF = 0.24$), while the result for estimated replicability was inconclusive (one-tailed $BF = 0.73$; two-tailed BF = 0.40).

The exploratory analyses of the remaining two dependent variables showed a similar pattern for trust in the company that developed the new treatment method. There was an interaction effect ($BF > 100$) due to decreased trust when authors revealed a conflict of interest in the positive outcome condition ($BF > 100$) and no effect of conflicts of interest when the results were negative (one-tailed $BF = 0.19$, two-tailed $BF = 0.17$). For trust in the journal, the means were in the same direction, but the mean differences were substantially smaller so that the result of the interaction test remained inconclusive (BF = 1.13).

Again, we report the simple effects of study outcomes for the sake of completeness. For the condition with conflicts of interest, one-tailed *t*-tests showed a substantial decrease in trust for all five measures (all *BF*s > 100). However, the two-tailed test of the effect of study results absent conflicting interests was conclusive only for estimated replicability (*BF* = 0.23). The remaining BFs varied between 0.54 and 1.14.

The results of Experiment 2 suggest that the findings from Experiment 1 are unlikely to be specific to the context of drug effectiveness tests. In an entirely different domain, we observed the same pattern, an interaction between conflict of interests and study outcome due to a trust discount when researchers with a COI reported positive results, but no change in trust for publishing findings that go against the authors' conflicting interests.

One possible explanation for the absence of the hypothesised trust boost is that the lay participants in Experiments 1 and 2 had limited knowledge of the potential costs to the researcher of going against a funder's interests. We test this possibility in Experiment 3 by running a replication of Experiment 1 with a sample of published scientists.

Experiment 3

Participants and design

Experiment 3 was based on the same two-factorial design as Experiment 1. Our initial preregistered plan was to recruit colleagues from our own institution as participants. Our aim was to recruit as many participants as possible in a one-month period. Similar to Experiment 2, we also preregistered that we would gather additional data in case the initial results were inconclusive and that we would recruit participants outside our institution if necessary. Instead of offering participants £0.50 as compensation, we entered them into raffles for £50 shopping vouchers (we allotted one voucher for each complete set of 50 participants). Since uptake within our own institution and neighbouring institutions was low ($N = 75$), and results were inconclusive, we decided to recruit additional participants via Prolific, who we compensated with £0.50. Recruiting additional participants via Prolific allowed us to increase the sample to

292 participants We had to exclude 11 participants who did not provide consent to take part and another 28 for failing the attention check. As we restricted participation to those who held a PhD in science, we excluded a further 81 participants whose PhD was not in a basic or applied science and another 11 who stated that they had not yet published empirical work. This left us with a total sample of 161 participants.

Procedure

The procedure of Experiment 3 was identical to that of Experiment 1 with one exception. After responding to the attention check, participants were asked to answer two additional questions in a yes-no format. First, they stated whether they held a PhD in a basic or applied science. Second, they indicated whether they had previously published empirical work.

Results

We again first tested the interaction effect of conflict of interest and study results, finding evidence for an interaction on trust in the study ($BF = 21.22$) and trust in the authors ($BF = 20.13$), but the result was inconclusive for estimated replicability ($BF = 1.17$) despite the pattern of means being qualitatively similar to the other two measures. We proceeded to disentangle the interaction effects by testing the simple effects of conflicts of interest on trust in the study and the authors. Again, trust decreased because of declared conflicts of interest when the results were positive (both BFs > 32.47). When the results were negative, trust in the study was unaffected by conflicts of interest (one-tailed $BF = 0.30$, two-tailed $BF = 0.31$), while the result was inconclusive for trust in the authors (one-tailed $BF = 0.71$, two-tailed $BF = 0.44$).

Exploratory analyses for trust in the journal and trust in the company that had developed the drug yielded inconclusive evidence regarding the predicted interaction effect, $BF = 0.61$ and $BF = 2.13$, respectively. Finally, one-tailed t-tests yielded evidence for a simple effect of study result when authors reported a conflict of interest for all trust measures (all

*BF*s > 12.01) except trust in the journal (*BF* = 1.31). The two-tailed *t*-tests of the corresponding simple effect when conflicts of interest were absent only yielded conclusive evidence against an effect of study results for estimated replicability (*BF* = 0.31). For the remaining measures, the result was inconclusive with BFs ranging between 0.34 and 0.62. These and other focal tests may have remained inconclusive due to the substantially smaller sample size in this Experiment than in Experiments 1 and 2.

The general pattern of results in Experiment 3 replicates those of the previous experiments. That is, even when restricting the sample to participants who should have some understanding of the potential costs of going against the interests of a funder, we only find a loss of trust when study results align with researchers' conflicting interests, but no trust boost for those who publish results that go against these interests. Thus, the results of Experiment 3 cast some doubt on our alternative explanation that a trust boost should occur once people understand the potential costs to researchers who publish findings that misalign with their funders' interests. In Experiment 4 we tested our alternative explanation in a different way by ensuring that participants understood the potential costs to researchers of publishing negative results.

Experiment 4

Participants and design

Experiment 4 is based on the same two-factorial design as the previous experiments. We aimed for a sample of 400 participants based on the same considerations as in Experiment 2. Participants were 449 UK residents recruited via Prolific. We had to exclude 61 participants due to failing the attention check, leaving us with a final sample of 388 participants.

Procedure

The procedure of Experiment 4 is identical to that of Experiment 1 with one exception. In the brief section preceding the title page of the fictional journal article, we added some information about conflicts of interest in science. Specifically, we told participants that

journals often require authors to disclose those conflicts, and that a typical conflict of interest was receiving funding from a company for empirical studies testing the effectiveness of its products. We further outlined that in such cases, scientists might be more motived to report results showing the effectiveness of the product because this might lead to continued funding. In contrast, reporting findings that speak against the effectiveness of the product could lead to funding being discontinued. This additional information aimed to make positive and negative payoffs to researchers contingent on their decisions about publication transparent to participants.

Results

We first tested for the interaction of conflicts of interest and study results on the three facial dependent variables. In line with Hypothesis 1, this interaction effect was present for trust in the study ($BF > 100$) and trust in the authors ($BF = 81.29$), but the evidence was inconclusive for estimated replicability ($BF = 0.97$). The inspection of the simple effects underlying the observed interaction effects showed that, consistent with the previous experiments, reporting a conflict of interested reduced trust in the study and the authors when the study results were positive (both BFs > 100). With negative results, disclosing a conflict of interest did not increase trust in the study. The preregistered one-tailed tests yielded inconclusive results for both trust in the study ($BF = 0.48$) and in the authors (BF = 0.37). However, since the pattern was very similar to those obtained in the previous experiments and the two-tailed test showed evidence for the absence of an effect for both measures (both $BF <$ 0.31), we accepted this finding as the final piece of evidence against the trust boost we predicted in Hypothesis 3. For the sake of completeness, we also explored the simple effects for estimated replicability, finding that a conflict of interest decreased estimated replicability when the study results were positive ($BF > 100$). For negative results, the result of the simple effect test yielded conclusive evidence against a trust premium (one-tailed $BF = 0.08$). While the subsequent two-tailed test yielded inclusive evidence, this was due to estimated

replicability being slightly lower when a conflict of interest was disclosed ($BF = 1.02$), which further speaks against Hypothesis 3.

Exploratory analysis of the two remaining trust measures yielded evidence for an interaction effect of conflict of interest and study results (both BFs > 6.75). For positive results, disclosing a conflict of interest reduced trust in both the journal that published the study ($BF = 6.10$) and the company that produced the drug (BF > 100). For negative results, trust in the company that produced the drug remained unchanged (one-tailed $BF = 0.33$; two-tailed $BF = 0.24$), while the test remained inconclusive for trust in the journal (one-tailed $BF = 1.16$, two-tailed $BF = 0.63$).

We continued with an exploratory inspection of the simple effects of study results. Without a conflict of interest, the results of the fictional study had no impact on trust in the study itself ($BF = 0.24$), in the journal ($BF = 0.22$), or on estimated replicability ($BF = 0.22$), while the result for trust in the authors was inconclusive ($BF = 0.44$). Surprisingly, trust in the company that developed the new drug was substantially lower when the results were negative ($BF = 9.34$), perhaps because participants thought that a company that produces ineffective drugs should not be trusted. When a conflict of interest was declared, positive results led to reduced trust on all measures except trust in the company (all BFs > 15.71). While the means were in the same direction, the result for trust in the company was inconclusive ($BF = 0.75$). In sum, the findings of Experiment 4 support Hypotheses 1 and 2 but provide evidence against Hypothesis 3.

The results of Experiment 4 resemble those of the three previous experiments. Even though we made the possible costs of publishing findings that go against a funder's interest as salient as possible, trust in the study or its authors and estimated replicability did not increase when the authors published a finding that clearly went against the conflicting interest. That is, even under what we consider ideal conditions for observing a trust boost for going against a funder's interests in empirical science, this trust boost did not manifest. Instead, we find the

familiar pattern where trust in science is affected only when the study result aligns with the potential conflict of intertest.

Experiment 5

Experiments 1 to 4 consistently showed an asymmetry in how conflicts of interest affect trust in science. When results are positive, thus aligning with the conflicting interests, trust is reduced. When results are negative, it remains at the level encountered when no conflict of interest is disclosed, that is, there is neither a trust loss nor a trust premium for researchers publishing findings that go against their conflicting interests. In Experiment 5, we tested two possible explanations for our findings. An attribution account which states that, per default, consumers of science attribute the published findings to honest reporting of the study results. Disclosure of conflicting interests now offers an alternative attribution of the published findings if those findings align with the conflicting interests. In contrast, findings that go against the conflicting interests cannot be plausibly attributed to dishonest reporting. The alternative is a more complex cue-based approach, in which disclosed conflicts of interest serve as a negative cue to research integrity irrespective of the study results. At the same time, the self-sacrifice of going against the conflicting interests, serve as a positive cue for research integrity (Sah & Feiler, 2020). Importantly, because the self-sacrifice is proportional to the temptation produced by the conflicting interests, the trust loss incurred due to disclosing a conflict of interest could be fully compensated by the trust gain that results from the self-sacrifice of publishing negative results despite an incentive not to.

Both accounts would predict the pattern we found in the first set of experiments. However, they make differing prediction when study authors disclose a conflict of interest that can be attributed externally. As Sah and Feiler argue – and show – in the context of medical advice, external attribution undermines the validity of conflicting interests as a negative cue while the validity of the self-sacrifice remains unchanged. Thus, if the cue-based approach holds true, externally induced conflicts of interests should not produce a trust loss

when results are positive and should even produce a trust premium for positive results. For the attribution account, in contrast, the origin of the conflict of interest should not matter. That is, externally induced conflicts of interests should have the exact same effects as regular conflicts of interest. In Experiment 5, we added a condition with externally induced conflicts of interest to subject these mutually exclusive accounts to an empirical test.

Participants and Design

Experiment 5 is based on a 2 (*conflict of interest*: conflict vs. no conflict vs. external conflict of interest) × 2 (*study outcome*: positive vs. negative) between-subjects design. We aimed for a sample of 600 participants for consistency with Experiment 4. Participants were 600 UK residents recruited via Prolific. We had to exclude 83 participants due to failing the attention check, leaving us with a final sample of 517 participants.

Procedure

The procedure of Experiment 5 was similar to that of Experiment 1 with one exception. We added externally induced conflicts of interest a third level of the conflict of interest factor. In this condition, the study authors declared that they had received funding from the manufacturer of the new drug via an existing agreement that had been negotiated by the retired former head of their institution. The disclosure statement added that the authors themselves did not negotiate the agreement nor did they have any influence on it being negotiated.

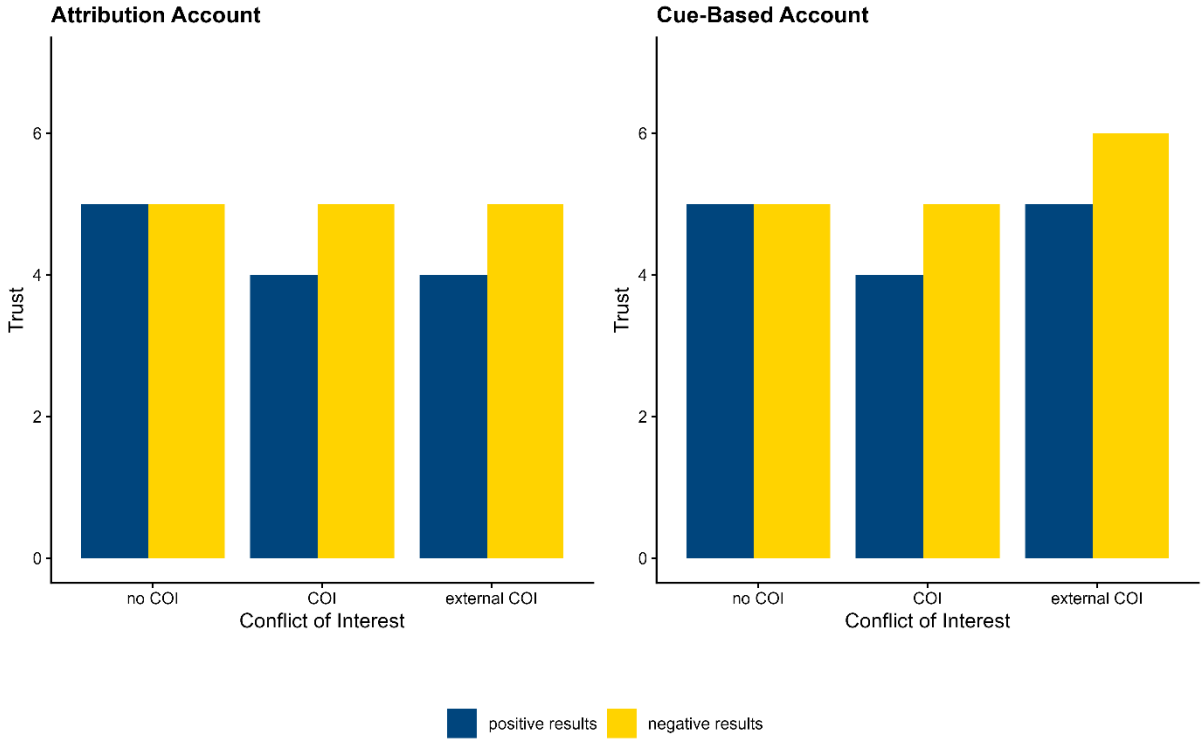
Results

We analysed our data using Bayesian Informative Hypotheses tests as implemented in the BAIN packages for R (Gu et al., 2019). Based on the two theoretical accounts, we specified the expected pattern of means for all focal trust measures in our 3 × 2 design as follows (see also Figure S1 below):

Attribution Hypothesis: negative/no COI = negative/COI = negative/external COI = positive/no COI > positive/COI = positive/external COI

Cue Hypothesis: negative/external COI > negative/no COI = negative/COI = negative/external COI = positive/no COI = positive/external COI > positive/COI

Figure S1. *Predicted Mean Trust Ratings for Experiment 5 by Theoretical Account*



In the BAIN analysis, we compared the two hypotheses with each other and with an unconstrained model, which contains all remaining mean patterns. Each hypothesis is evaluated against this unconstrained model, yielding a Bayes factor. Since Bayes factors a re transitive, the two competing hypotheses can be compared by computing a Bayes factor as the ratio of two Bayes factors relative to the unconstrained model. Similar to the previous experiments, our threshold for accepting a hypothesis as true was that its BFs exceeded 3 when compared to both the unconstrained model and the competing hypothesis. In addition, the BAIN package computed the posterior probability of each hypothesis (including the unconstrained model) being true. We defined a posterior probability of at least .90 as an additional threshold for accepting a hypothesis as true.

The attribution hypothesis explained the data better than both the cue hypothesis and the unconstrained model for trust in the study, trust in the authors, and expected replicability (all BFs >100). The posterior probability of the attribution hypotheses being true exceeded .99 for trust in the study and trust in the authors, and it exceeded .98 for estimated replicability. Not surprisingly given this clear pattern of results, an externally induced COI produced essentially the same effects as a regular COI disclosure, that is, trust was lost and replicability estimated to be lower when results aligned with the externally induced COI, but there was neither a trust loss nor trust gain in case of negative study results.

Similar to the previous experiments, we subjected trust in the journal and trust in the manufacturer of the new drug to the same BAIN analysis for exploratory reasons. The results for the two additional trust measures mirror those of the confirmatory tests. That is, trust in the journal as well as trust in the funder was explained better by the attribution hypothesis than the other models (BFs > 100), and the posterior probabilities of the attribution hypothesis exceeded .98 in both cases.

In sum, Experiment 5 strongly supports the attribution explanation of trust in science. Irrespective of the exact nature of a disclosed conflict of interest, participants considered it in the light of the published findings. Their trust in science was reduced only when the existence of conflicting interests offered a plausible alternative explanation to honest reporting. Otherwise, trust was unchanged, which also means that, once again, there was evidence against a trust premium for researchers publishing findings that go against their competing interests.

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